

### **PROBLEM C: Nonrenewable Resources**

Select a vital nonrenewable or exhaustible resource (water, mineral, energy, food, etc.) for which your team can find appropriate world-wide historic data on its endowment, discovery, annual consumption, and price.

The modeling tasks are:

1. Using the endowment, discoveries, and consumption data, model the depletion or degradation of the commodity over a long horizon using resource modeling principles.
2. Adjust the model to account for future economic, demographic, political and environmental factors. Be sure to reveal the details of your model, provide visualizations of the model output, and explain limitations of the model.
3. Create a fair, practical "harvesting/management" policy that may include economic incentives or disincentives, which sustain the usage over a long period of time while avoiding severe disruption of consumption, degradation or rapid exhaustion of the resource.
4. Develop a "security" policy that protects the resource against theft, misuse, disruption, and unnecessary degradation or destruction of the resource. Other issues that may need to be addressed are political and security management alternatives associated with these policies.
5. Develop policies to control any short- or long-term "environmental effects" of the harvesting. Be sure to include issues such as pollutants, increased susceptibility to natural disasters, waste handling and storage, and other factors you deem appropriate.
6. Compare this resource with any other alternatives for its purpose. What new science or technologies could be developed to mitigate the use and potential exhaustion of this resource? Develop a research policy to advance these new areas.

### **ICM2005 - C 题: 不可再生资源管理**

选择一种必不可少的不可再生的或会枯竭资源，比如水、矿藏、能源、食物等等，对此你们队能找到合适的世界范围内有关其天赋资源、已知的蕴藏量、年消耗量、价格等历史数据。建模的任务为：

- 1、利用天赋资源、发现、年消耗量、价格等历史数据，运用资源建模的原则，建立这种不可再生资源在长时间中逐渐耗尽或退化的模型。
- 2、考虑未来的经济、人口统计学、政治和环境等因素来调整模型。务必展示你们模型的细节，提供模型结果直观形象的表示，并指出模型的局限性。
- 3、创建一种公平、实用的“开采/ 管理”策略，其中要包括经济上的激励和抑制措施，从而确保资源的长期可用性，而且能避免这种资源的极度混乱的消耗、退化或者迅速枯竭。
- 4、研制一种“安全保障”策略以保护资源被偷盗、滥用、遭到破坏、不必要的退化或毁灭。可能需要讨论问题有与这些策略相应的政策和安全管理方面的候选方案。
- 5、研制控制开采的短期或长期“环境影响”的策略。务必包括诸如污染物、不断增长的对自

然灾害的敏感性、垃圾处理和存储，以及其他你们认为重要的因素。

6、把这种资源和相近用途的其他资源进行比较，为减少你选择的资源的消耗或潜在的耗尽危险应该发展什么样的新的科学或技术？研制一种能推动这些新领域的进步的研发策略。